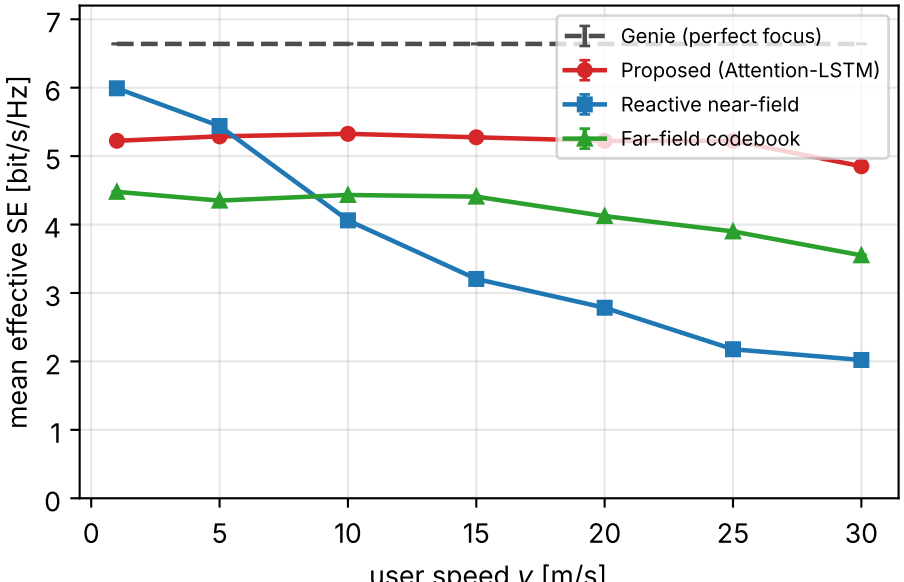
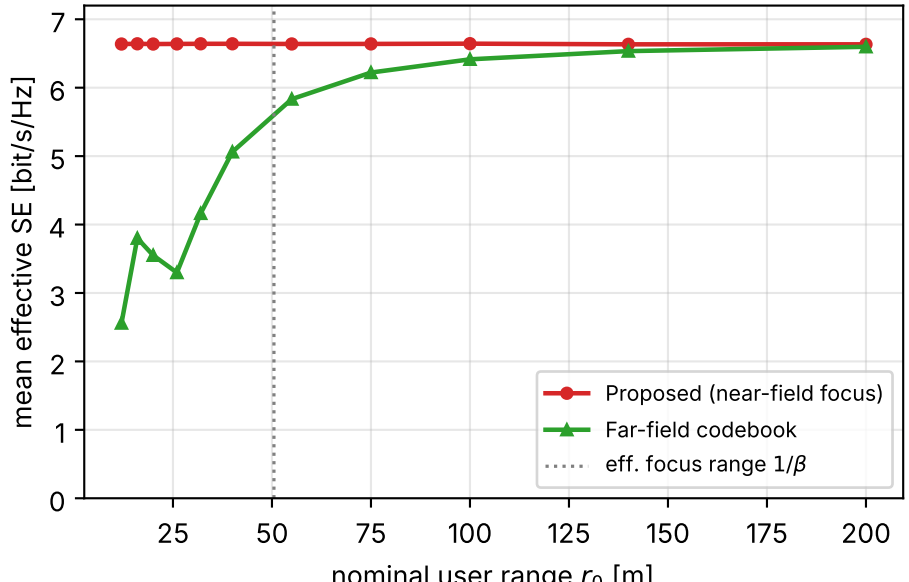

(a) Robustness to mobility (trained predictor)
(τ_{lat}=20 ms, $r_0 \approx$ 20 m, B=4)
Genie (perfect focus)
Proposed (Attention-LSTM)
Reactive near-field
Far-field codebook
mean effective SE [bit/s/Hz]
user speed v [m/s]
0
1
2
3
4
5
6
7
0
5
10
15
20
25
30


(b) Near-field focusing advantage vs range
(v=15 m/s)
Proposed (near-field focus)
Far-field codebook
eff. focus range $1/\beta$
mean effective SE [bit/s/Hz]
nominal user range r_0 [m]
0
1
2
3
4
5
6
7
25
50
75
100
125
150
175
200